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ELI5 Data center power and water use

Posted by u/pluk78 • • 0 points • 40 comments

It's a rare thing to find any article about the general horrors of AI that does not mention the water and power usage destroying the earth.

Over simplisticly, for many variants of power production the energy is spent to create heat that drives turbines in one way or another.

If these AI machines are creating so much heat, and we pour on water to cool them to create steam, why can't that steam be used to generate power, and the water condense back down effectively in a close system? There would obviously still be usafe of both, but it feels like that would mitigate a lot of the damage being done?

Comments

u/explainlikeimfive-ModTeam • 1 points • 2026-02-05 00:34 UTC

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u/white_nerdy • 1 points • 2026-02-04 15:27 UTC

One interesting thing other commenters miss is that demand for energy is increasing demand for alternative technologies. For example they're [building a power plant](https://www.reuters.com/business/energy/helion-energy-starts-construction-nuclear-fusion-plant-power-microsoft-data-2025-07-30/) to power AI datacenters with fusion (!) energy.

If a few years of higher electric bills incentivize the free market to develop a means to permanently make electricity cheap and abundant, overall it's a net gain for humanity.

u/MrWigggles • 1 points • 2026-02-04 14:00 UTC

Steam turbines need high pressured steam.

While datacenters do get hot, they do not stay constantly hot beyond the boiling temp of water. Which is what happens in any steam turbine boiler.

u/fiendishrabbit • 1 points • 2026-02-04 08:30 UTC

Note that countries with more stringent waste heat management laws often do use the excess water from data centers for something useful.

It's not hot enough to be used to generate electricity, but it can be used to for example heat a public swimming pool or even district heating.

u/cyann5467 • 1 points • 2026-02-04 07:05 UTC

The issue with water usage isn't that the water is destroyed. It's that any given area only has so much water that can be used at any given time. Data centers are using so much water that the surrounding areas don't have any to use themselves.

This video goes into it in more detail.

https://youtu.be/H_c6MWk7PQc?si=cMB6qhcunGk1JYz6

u/jsher736 • 1 points • 2026-02-04 01:19 UTC

You can recycle it but it's not a perfect process so you always lose some water

u/tylerlarson • 1 points • 2026-02-03 23:15 UTC

Water really isn't the big problem. Of all the ways water gets used, cooling data centers is a comparatively very environmentally friendly use and even in the most extreme scenarios it's not a lot of water anyway.

The big problem is still power consumption. Hell, even water use is technically still a matter of power consumption, since getting more clean water just takes energy.

But talking about water consumption seemed to strike a nerve, so it's become a major talking point.

> **u/mtbdork** • 0 points • 2026-02-04 01:00 UTC
>
> Energy required for pumping water increases at a cubic rate with flow. As more data centers come online, energy for both water and power means our energy bills go up. A lot.

u/paulHarkonen • 1 points • 2026-02-03 22:13 UTC

The underlying process is very straightforward. Data center chips use electricity to do math. Doing math is really hard work so they heat up just like you do when you have to work hard. In order to keep the chips from overheating they essentially pump water over the chips. But that water is now hot so they take that water and spray it over a fan to cool it off. Some of the water evaporates, so they need to get more water to replace it. Multiply that system millions of times over and you have a system that uses a lot of water and a lot of electricity.

Data centers equipment don't get boiling hot, they get hot tub hot (ideally they don't even get that hot and are closer to a warm bath hot). If they get to boiling water hot, they shut down due to overheating as that damages a lot of systems. Warm bath water isn't hot enough to make electricity so you can't really recover it. There are closed loop cooling systems, but they're much less efficient than evaporative cooling (it's the difference between how hot a hot bath feels vs a warm shower in 5 year old terms) so data centers use evaporative systems.

u/[deleted] • 1 points • 2026-02-03 20:45 UTC

[removed]

u/vankirk • 1 points • 2026-02-03 20:44 UTC

Even though you can't do this with data centers, this is exactly what a combined cycle electric generator does. You burn NG like a jet engine that spins a turbine and creates electricity, then you use the steam produced through the NG combustion to turn a steam turbine.